

Day 2 – Sampling Controls: Temperature, Top-k, and Top-p (Summary with Examples)

1. The Basic Idea

When an LLM generates text, it first produces a list of possible next tokens with probabilities. Temperature, top-k, and top-p do not change what the model knows; they only control how the next token is chosen from that probability list. Together, they decide how strict, safe, or creative the output will be.

2. Temperature (How Random the Choice Is)

Temperature reshapes the probability distribution. At low temperature (0.1–0.3), the highest-probability token becomes even more dominant, so the model almost always picks the top choice. This makes outputs more deterministic and reliable, which is ideal for code, SQL, and factual tasks. At medium temperature (0.5–0.7), the model sometimes picks the second or third best option, making answers more natural for explanations. At high temperature (0.9+), lower-probability tokens become more likely, increasing creativity but also the risk of errors and hallucinations.

3. Top-k Sampling (Limit by Count)

Top-k means: keep only the K most likely tokens and discard the rest. For example, if top-k = 2, the system keeps only the top two tokens by probability and removes all others. Then the remaining probabilities are renormalized, and one token is sampled from this smaller set. Smaller k values make output safer and more predictable; larger k values allow more variety and risk.

4. Top-p (Nucleus Sampling) (Limit by Probability Mass)

Top-p means: keep the smallest set of tokens whose total probability is at least p (for example, p = 0.9). The system starts from the most likely token and keeps adding tokens until their cumulative probability reaches the chosen threshold. Then it renormalizes and samples from only that set. If the model is very confident, top-p may keep only one or two tokens. If the model is uncertain, it may keep many tokens. This makes top-p adaptive to the situation.

5. How They Work Together

In practice, generation usually works like this: (1) the model outputs probabilities, (2) temperature reshapes those probabilities, (3) top-k or top-p removes unlikely options, and (4) one token is sampled from what remains. Temperature controls how adventurous the choice is, while top-k and top-p control which options are even allowed to be considered.

6. Practical Rules of Thumb

- 1 For code and factual tasks: use low temperature (0.1–0.3) and a conservative top-k or top-p (e.g., top-p = 0.8–0.9).
- 2 For explanations and chat: use medium temperature (0.5–0.7) with top-p around 0.9.
- 3 For creative writing and brainstorming: use higher temperature (0.8–1.0+) with top-p around 0.95.
- 4 Temperature controls randomness; top-k limits by number of options; top-p limits by total probability mass.

- 5 All three together balance reliability versus creativity and help reduce hallucinations when set conservatively.